Subject: Muhammed Ali KAYA - 150719067 Date: ...../...../.....

"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh dixipline issues".

1)

$$A=\begin{bmatrix} 1 & \alpha & -y & \alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \quad \begin{matrix} S_2+S_3 \to S_3 \\ \frac{-S_1}{2}+S_4 \to S_4 \end{matrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \quad \frac{1}{2}+k=0 \quad \boxed{k=\frac{-1}{2}}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} 8S_1+S_2 \to S_2 \\ \frac{-S_3}{2}+S_4 \to S_4 \end{matrix} \quad \boxed{m=-2}$$

b)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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2) $x = 5y \Rightarrow v_1 = \begin{bmatrix} 5y \\ y \end{bmatrix} = y\begin{bmatrix} 5 \\ 1 \end{bmatrix}$ $\quad C_1 \cdot V_1 = 0$

*2 dimensen space

c) $x = 6 - 4y + z$ $\quad \begin{bmatrix} -4y+z \\ 4y \\ z \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -4 & 1 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -\lambda & -4 & 1 \\ 0 & 4-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix}$ *3 dimensional space.

*These vectors from vector space.

$\lambda_1 = 1$ , $\lambda_2 = 4 \Rightarrow$

3) a) $H = I - 2u \cdot u^T \Rightarrow H^2 = (I - 2u \cdot u^T)^2 = I^2 - 4uu^T + 4uu^T u \cdot u^T$

$u^T u = 1$, $H^2 = I$ $\quad H = H^T$, $H \cdot H^T = 1 \Rightarrow$

b) implying that H is an orthogonal matrix.

c) $Hu = (I - 2u \cdot u^T) \cdot u = u - 2u \cdot u^T u = u - 2u = -u$, $\lambda = -1$

d) Since ith diagonal entry of $uu^T$ is $u_i^2$, the i diagonal entry of H is $H_{ii} = 1 - 2u_i^2$

$\sum_{i=1}^{n} H_{ii} = n - 2 \sum_{i=1}^{n} u_i^2 = n - 2$. H also $\sum \lambda_i = 1 - 1 + (n-1) \cdot 1$ $\quad$ " $n-2$

4) First tank $V(0) = 10L$ , Second tank $w(0) = 40L$

a) $\begin{bmatrix} V_{k+1} \\ W_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} V_k \\ W_k \end{bmatrix} \Rightarrow \begin{bmatrix} -2-\lambda & 1 \\ 1 & -2-\lambda \end{bmatrix}$

$(-2-\lambda)^2 - 1 = 0 \Rightarrow \lambda_1 = -3$ , $\lambda_2 = -1$

$\lambda_1 = -3 \Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ , $\lambda_2 = -1 \Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \cdot x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} c_1 + c_2 = 1 \\ -c_1 + c_2 = 0 \end{matrix} \Rightarrow c_1 = c_2 = 1/2$

$t = 1 \Rightarrow$ $u_1 = \frac{(-3)}{2} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} - \frac{1}{2} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} =$

$u_t = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot (-3)^t \cdot \frac{1}{2} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-1)^t \cdot \frac{1}{2} \Rightarrow$

$u_1 = \begin{bmatrix} -3/2 \\ 3/2 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$.

$t \to \infty \Rightarrow u_\infty = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ %50 of tank 1

%50 of tank 2

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